

RICHMOND, CA, USA

WMO: 994036

Lat: 37.923N

Lon: 122.410W

Elev: 18

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 42.2	(c) 43.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 22.5	(k) 56.2	(l) 20.1	(m) 54.1	(n) 6.5	(o) 350	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 10.5	(c) 77.0	(d) N/A	(e) 73.7	(f) N/A	(g) 71.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.2	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.2	13.7	DB	39.3	86.3	2.2	4.8	37.7	89.7	36.4	92.5	35.1	95.2	33.5	98.7	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	52.2	53.3	55.2	56.9	58.5	60.8	61.5	62.3	64.0	62.1	57.0	51.6
	DBStd	5.53	3.84	3.89	3.77	4.03	3.53	3.62	3.00	3.31	3.80	4.25	3.75	3.99
	HDD50	63	23	10	2	0	0	0	0	0	0	0	1	27
	HDD65	2661	396	328	305	247	206	136	117	97	61	110	243	415
	CDD50	2975	92	102	162	207	264	324	356	383	421	377	210	77
	CDD65	96	0	0	0	3	5	10	8	14	32	22	2	0
	CDH74	268	0	0	1	21	23	48	12	24	72	63	4	0
	CDH80	45	0	0	0	1	7	13	0	2	13	9	0	0
Wind	WSAvg	6.5	6.6	6.7	6.4	7.2	7.2	7.1	7.1	6.6	5.6	5.3	5.8	6.2
Precipitation	PrecAvg	24.6	5.1	4.9	2.7	1.6	0.9	0.2	0.0	0.1	0.0	1.2	2.4	5.4
	PrecMax	40.2	13.3	16.0	9.2	5.9	3.6	0.9	0.0	0.9	0.3	4.6	7.7	13.4
	PrecMin	11.9	0.3	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	PrecStd	7.2	3.7	3.4	2.7	1.4	0.9	0.2	0.0	0.2	0.1	1.2	2.2	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.6	69.6	71.2	77.3	76.4	81.6	75.5	77.3	81.1	80.1	73.1	64.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	62.9	64.1	65.9	69.8	69.9	72.9	71.1	72.8	75.3	75.1	68.1	61.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	59.8	61.0	63.0	65.7	66.2	69.2	68.6	70.0	72.2	71.0	64.3	59.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	57.4	58.9	60.4	62.4	63.8	66.5	66.4	67.9	69.4	67.8	62.0	57.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	9.0	8.9	9.7	10.6	10.0	10.6	9.6	9.7	10.5	10.1	9.1	7.7
		MCDBR	12.8	12.7	13.7	15.3	14.8	15.6	12.6	12.9	14.5	15.5	12.9	9.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.322	0.327	0.338	0.357	0.362	0.357	0.365	0.345	0.344	0.333	0.321	0.323	
	taud	2.560	2.546	2.505	2.420	2.435	2.454	2.433	2.490	2.481	2.555	2.566	2.561	
	Ebn at Noon	274	288	294	292	291	292	288	293	287	282	272	266	
	Edh at Noon	24	27	31	36	36	36	36	33	32	27	23	23	
All-Sky Solar Radiation	RadAvg	749	1045	1477	1940	2246	2457	2336	2078	1757	1286	883	653	
	RadStd	108	131	137	155	145	108	79	88	66	86	59	98	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (46 neighbors)		N/A	N/A	-2.14	N/A	N/A	N/A	N/A	-198	+243	N/A

Nomenclature: See separate page